

C4: Yingchuan parameter checkpoint lambda=0.7, beta=0.0005

Sapelo array 48497633; five prespecified exploratory seeds. This is one of eight sequential checkpoints. All five jobs are terminal before this report; no next beta was launched for its construction.

Evidence boundary: measured CIFAR-100-C severity-5 trajectories and finite intervention responses. Numerical failure is a failed computation, not behavioral deterioration. A finite F response does not identify a closed Jacobian, spectrum or phase transition.

Complete five-seed status

Seed	Parent blocks	Extension blocks	Forks verified	H2O analyses	Coverage / reason
202	15	30	4/4	2/2	complete
303	15	30	4/4	2/2	complete
404	15	30	4/4	2/2	complete
505	15	30	4/4	2/2	complete
606	15	30	4/4	2/2	complete

A component is reported only after its own completion and verification. Parent, extension, fork and H2O gaps remain explicit; the raw run and failed attempts remain on Sapelo.

Exploratory readout: visit-3 minus visit-1 same-domain CE is positive in 2/5 seeds (mean -0.025); classification error changes by -4.91 percentage points on average. At H1410, 4/10 correlated parent nodes have harmful closed acceptance and 5/10 have positive feedback interaction. The signs are mixed; these are descriptive, not a persistent-deterioration or stability claim.

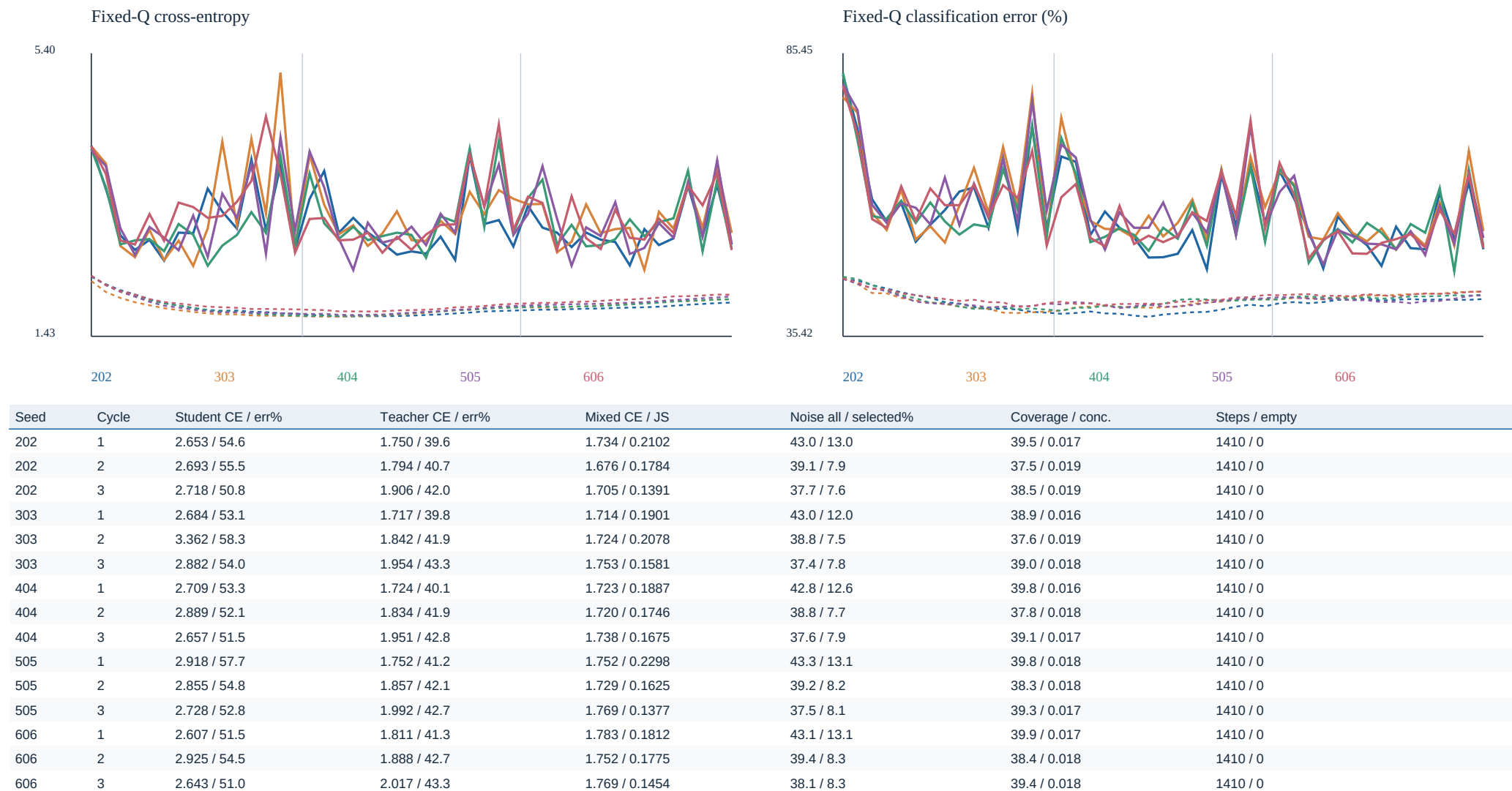
C4-1. Proxy versus labelled local update

Negative post-minus-pre CE means local improvement. Near zero means either absolute change is below 1e-4. Natural C4 executes updates without a separate accept/reject evaluator.

Seed	Probes	Both improve	Proxy down / local up	Proxy up / local down	Both worsen	Near zero
202	90	58	32	0	0	0
303	90	49	40	0	1	0
404	90	47	42	0	1	0
505	90	52	38	0	0	0
606	90	45	44	0	1	0

C4-2. Complete fixed-Q trajectories and pseudo-supervision

The fixed-Q panel uses the same 1,500 image-corruption evaluations at every block. Curves are unsmoothed. Dotted curves are teachers; solid curves are students. Missing/failed cells are absent, not assigned zero risk.



Cycle CSV retains teacher/student disagreement, mixed-target confidence, count-weighted pseudo-label statistics, selected-class concentration and elapsed block time. Selected noise is undefined when no examples were selected.

C4-3. Same-domain returns

Visit-3 minus visit-1 post-block student CE for each named corruption. Positive means worse on that domain; all domains and all terminal cells are retained. The companion block CSV includes pre/post CE and error for student and teacher at every visit.

Corruption	202	303	404	505	606
gaussian_noise	-0.197	-1.474	-0.662	-0.631	-0.845
shot_noise	-0.913	-0.494	0.756	0.717	-0.644
impulse_noise	0.279	0.039	0.211	0.345	0.094
defocus_blur	0.239	0.185	0.324	0.105	0.400
glass_blur	-0.277	0.215	-0.110	-0.308	-0.150
motion_blur	0.388	0.278	0.212	0.253	0.221
zoom_blur	0.174	0.204	0.167	0.341	0.218
snow	-0.400	0.458	0.451	-0.090	0.031
frost	-0.876	-0.617	0.544	0.238	0.271
fog	-0.097	-0.222	0.182	-0.248	0.140
brightness	0.077	0.270	0.340	0.152	-0.012
contrast	-0.255	-0.466	-0.186	0.556	0.132
elastic_transform	0.443	0.123	0.191	0.083	-0.311
pixelate	-1.030	-1.429	-0.862	0.409	-0.113
jpeg_compression	0.138	0.132	0.138	0.174	-0.015

C4-4. Per-seed parameter summary

Seed	Mean post CE / err%	Mean return dCE / dErr pp	Cycle-3 CE / err%	Coverage%	Updates	Block cost min
202	2.615 / 51.8	-0.154 / -6.20	2.552 / 49.1	38.5	4230	17.5
303	2.602 / 51.2	-0.186 / -6.59	2.539 / 48.5	38.5	4230	17.7
404	2.602 / 51.4	0.113 / -3.30	2.644 / 50.0	38.9	4230	17.5
505	2.692 / 51.9	0.140 / -4.37	2.734 / 49.6	39.1	4230	16.6
606	2.592 / 50.7	-0.039 / -4.10	2.597 / 49.2	39.2	4230	16.6

C4-5. Matched accept/reject and closed/replay forks

The first nonempty update after preselected blocks 5 and 15 is the focal candidate. Positive accept-minus-reject CE is worse. $A_{\text{closed}} = \text{accept_closed} - \text{reject_closed}$; $A_{\text{open}} = \text{accept_replay} - \text{reject_replay}$; $I_{\text{feedback}} = A_{\text{closed}} - A_{\text{open}}$. The H94 and H1410 forks share their initial state and schedule; their latest common scored checkpoint, H93, agrees exactly. H94 is therefore a correlated branch, not an independent replicate. The H1410 run does not separately score H94.

Seed	Parent	Immediate Q dCE / dErr pp	Immediate domain dCE / dErr pp
202	5	-0.090 / -0.40	-0.054 / -0.27
202	15	-0.093 / 0.40	-0.265 / -2.33
303	5	0.015 / 1.40	-0.017 / 0.00
303	15	-0.025 / -0.13	-0.110 / -1.33
404	5	0.017 / 0.87	0.051 / 0.60
404	15	0.035 / 0.53	0.073 / 1.00
505	5	0.048 / -0.40	0.059 / -0.53
505	15	-0.017 / 0.40	-0.117 / -1.40
606	5	-0.021 / -0.13	-0.088 / -2.67
606	15	0.023 / 1.67	0.061 / 1.27

H94 fixed-Q CE contrasts

Seed	Parent	R closed	R replay	A closed	A replay	A_{closed}	A_{open}	I_{feedback}
202	5	2.462	2.462	2.516	2.475	0.054	0.013	0.040
202	15	4.379	4.379	3.469	3.652	-0.910	-0.727	-0.183
303	5	2.609	2.609	2.477	2.430	-0.132	-0.179	0.047
303	15	3.420	3.420	3.671	3.890	0.251	0.470	-0.219
404	5	2.565	2.565	2.618	2.571	0.053	0.007	0.046
404	15	3.612	3.612	3.703	3.694	0.091	0.082	0.009
505	5	2.742	2.742	2.913	2.812	0.171	0.070	0.101
505	15	4.003	4.003	4.175	4.108	0.173	0.105	0.068
606	5	2.874	2.874	2.868	2.988	-0.006	0.114	-0.119
606	15	2.927	2.927	3.069	3.092	0.142	0.165	-0.023

H1409 fixed-Q CE contrasts

Seed	Parent	R closed	R replay	A closed	A replay	A_closed	A_open	I_feedback
202	5	3.271	3.271	2.889	2.887	-0.381	-0.384	0.003
202	15	2.630	2.630	2.693	2.606	0.063	-0.024	0.087
303	5	2.834	2.834	2.701	2.862	-0.133	0.029	-0.162
303	15	3.234	3.234	3.362	2.942	0.129	-0.291	0.420
404	5	2.941	2.941	2.782	3.118	-0.159	0.177	-0.336
404	15	3.141	3.141	2.889	2.781	-0.252	-0.360	0.108
505	5	3.018	3.018	3.027	3.124	0.009	0.106	-0.097
505	15	2.830	2.830	2.855	2.822	0.025	-0.007	0.032
606	5	2.741	2.741	2.887	2.835	0.146	0.094	0.052
606	15	2.891	2.891	2.925	2.969	0.034	0.078	-0.044

H1410 fixed-Q CE contrasts

Seed	Parent	R closed	R replay	A closed	A replay	A_closed	A_open	I_feedback
202	5	3.353	3.353	2.812	2.835	-0.541	-0.518	-0.023
202	15	2.680	2.680	2.669	2.638	-0.012	-0.043	0.031
303	5	2.968	2.968	2.767	2.986	-0.200	0.018	-0.219
303	15	3.165	3.165	3.391	2.922	0.227	-0.243	0.470
404	5	2.942	2.942	2.850	3.189	-0.093	0.247	-0.339
404	15	3.148	3.148	3.043	2.844	-0.105	-0.304	0.199
505	5	3.054	3.054	3.100	3.182	0.046	0.128	-0.082
505	15	2.879	2.879	2.901	2.810	0.021	-0.070	0.091
606	5	2.733	2.733	2.926	2.844	0.194	0.111	0.083
606	15	2.938	2.938	2.907	2.931	-0.031	-0.007	-0.024

C4-6. Diagnostic state, finite F and local identifiability

x=(student fixed-Q CE, mixed-target fixed-Q CE, student/teacher JS, mixed confidence). F is the mean H20 future x over three base tapes at each observed parent; it is a conditional response, not a closed transition law. The block CSV stores every observed x_t .

Seed	Parent	x0	F_H20(x0)	Tape SD	Finite	J status
202	5	2.785 / 1.977 / 0.267 / 0.689	2.521 / 1.963 / 0.235 / 0.688	0.182 / 0.001 / 0.017 / 0.000	132/132	not identified
202	15	2.653 / 1.734 / 0.210 / 0.671	3.776 / 1.733 / 0.300 / 0.671	0.486 / 0.000 / 0.020 / 0.000	132/132	not identified
303	5	2.926 / 1.904 / 0.265 / 0.679	2.634 / 1.894 / 0.221 / 0.679	0.247 / 0.000 / 0.017 / 0.000	132/132	not identified
303	15	2.684 / 1.714 / 0.190 / 0.665	3.313 / 1.713 / 0.279 / 0.665	0.270 / 0.000 / 0.031 / 0.000	132/132	not identified
404	5	2.797 / 1.968 / 0.277 / 0.690	2.628 / 1.955 / 0.229 / 0.689	0.035 / 0.000 / 0.006 / 0.000	132/132	not identified
404	15	2.709 / 1.723 / 0.189 / 0.666	3.157 / 1.721 / 0.270 / 0.666	0.347 / 0.001 / 0.010 / 0.000	132/132	not identified
505	5	2.966 / 1.961 / 0.280 / 0.688	2.579 / 1.946 / 0.254 / 0.687	0.061 / 0.000 / 0.005 / 0.000	132/132	not identified
505	15	2.918 / 1.752 / 0.230 / 0.668	3.558 / 1.750 / 0.310 / 0.668	0.194 / 0.000 / 0.029 / 0.000	132/132	not identified
606	5	3.146 / 1.986 / 0.277 / 0.690	2.840 / 1.975 / 0.246 / 0.689	0.072 / 0.000 / 0.011 / 0.000	132/132	not identified
606	15	2.607 / 1.783 / 0.181 / 0.670	3.499 / 1.782 / 0.259 / 0.670	0.452 / 0.001 / 0.036 / 0.000	132/132	not identified

Perturbation, conditioning and held-out diagnostics

Seed	Parent	Median G	Rank	Condition	Scale diff	Held-out median	Closure	Reason
202	5	12.2	4	4.1	0.83	56.55	falsified	not established
202	15	86.8	4	4.7	0.33	353.44	falsified	not established
303	5	37.8	4	2.3	1.07	83.74	falsified	not established
303	15	34.9	4	5.4	0.88	159.08	falsified	not established
404	5	17.9	4	6.4	0.72	69.97	falsified	not established
404	15	62.9	4	6.2	1.21	211.91	falsified	not established
505	5	26.7	4	4.6	0.96	72.00	falsified	not established
505	15	60.8	4	4.9	0.72	188.94	falsified	not established
606	5	28.6	4	3.3	0.77	78.26	falsified	not established
606	15	72.7	4	33.9	0.86	376.94	falsified	not established

Independent student/teacher perturbations, input rank/condition, two scales, held-out directions and momentum closure checks are retained in the compact analysis files. No eigenvalue or phase boundary is claimed without a closed and predictive state law.

C4-7. Bounded parameter-map reading

This checkpoint isolates one pair $\lambda=0.7$, $\beta=0.0005$ with five exploratory seeds. Seeds are reused across later pairs for paired comparison; they are not independent cross-parameter replications. Outcomes at this pair alone cannot locate a stability boundary. Numerical failures and improvement/null findings remain in the reported denominator.

Observed coverage: 5/5 complete 45-block trajectories, 10 verified H1410 parent nodes, 4 positive A_{closed} and 5 positive I_{feedback} . Parent nodes within one seed are correlated. A positive endpoint or interaction alone does not establish sustained recursive deterioration.

Coverage ledger and missingness

C4 item	Present evidence	Missing / limitation
C4-1	450 completed local probes	Unavailable after failed/incomplete trajectories
C4-2	5 full trajectories and cycle process records	Failed/partial cells not extrapolated
C4-3	75 same-domain return comparisons	Only complete first/third visits
C4-4	5 per-seed parameter summaries	One pair cannot characterize a sweep
C4-5	30 fork horizons	Unavailable parent nodes explicitly excluded
C4-6	10 H20 parent analyses	J/spectrum only if closure and held-out criteria pass
C4-7	One-pair status/sign description	No identified region or boundary

Provenance: Sapelo 48497633, compact evidence MANIFEST.json and AUDIT.json. Raw checkpoints, logits, step tapes and all failures remain under /scratch/yz54720/RSI2RSD/C4/sweep_yingchuan_20260923/results/I07_b00005/. The reserved confirmation panel was not read.